# Matrices & Vectors

## Section 1: Matrix Multiplication & Inverses

**1** If $A$ is a $3 \times 2$ matrix and $B$ is a $2 \times 4$ matrix, what are the dimensions of the product matrix $AB$?

**a.** $3 \times 4$

**b.** $2 \times 2$

**c.** $4 \times 3$

**d.** Multiplication is undefined

**2** What is the inverse of the matrix $M = [[2, 5], [1, 3]]$?

**a.** $[[3, -5], [-1, 2]]$

**b.** $[[-3, 5], [1, -2]]$

**c.** $[[2, -5], [-1, 3]]$

**d.** $[[3, 1], [5, 2]]$

**3** Let $A$ and $B$ be invertible $n \times n$ matrices. Which expression represents $(AB)^{-1}$?

**a.** $B^{-1}A^{-1}$

**b.** $A^{-1}B^{-1}$

**c.** $AB^{-1}$

**d.** $BA^{-1}$

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## Section 2: Determinants & Cramer's Rule

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**4** What is the determinant of the $3 \times 3$ matrix $[[1, 2, 3], [0, 4, 5], [0, 0, 6]]$?

**a.** $24$

**b.** $30$

**c.** $36$

**d.** $0$

**5** If the determinant of a $2 \times 2$ matrix $A$ is $5$, what is $\det(3A)$?

**a.** $15$

**b.** $45$

**c.** $75$

**d.** $135$

**6** For what value of $x$ is the matrix $[[x, 3], [12, x]]$ singular?

**a.** $\pm 6$

**b.** $\pm 4$

**c.** $\pm 12$

**d.** $0$

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## Section 3: Dot Product & Angle Between Vectors

**7** What is the angle $\theta$ between vectors $\mathbf{u} = \langle 1, 0 \rangle$ and $\mathbf{v} = \langle 1, 1 \rangle$?

**a.** $30^\circ$

**b.** $45^\circ$

**c.** $60^\circ$

**d.** $90^\circ$

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**8** For what value of $k$ are the vectors $\mathbf{a} = \langle 4, k \rangle$ and $\mathbf{b} = \langle 3, -6 \rangle$ orthogonal?

**a.** $1$

**b.** $2$

**c.** $3$

**d.** $4$

**9** What is the scalar projection of vector $\mathbf{u} = \langle 4, 3 \rangle$ onto vector $\mathbf{v} = \langle 12, 5 \rangle$?

**a.** $\frac{63}{13}$

**b.** $5$

**c.** $\frac{48}{13}$

**d.** $\frac{15}{13}$

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## Section 4: 3D Vectors & Cross Product

**10** What is the cross product $\mathbf{i} \times \mathbf{j}$ in three-dimensional space?

**a.** $\mathbf{k}$

**b.** $-\mathbf{k}$

**c.** $\mathbf{0}$

**d.** $1$

**11** What is the magnitude of the vector $\mathbf{v} = \langle 2, -3, 6 \rangle$?

**a.** $7$

**b.** $9$

**c.** $\sqrt{49}$

**d.** $8$

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**12** What is the area of the parallelogram determined by vectors $\mathbf{a} = \langle 3, 0, 0 \rangle$ and $\mathbf{b} = \langle 0, 4, 0 \rangle$?

**a.** 6

**b.** 12

**c.** 24

**d.** 7

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